

# ANSWER SHEET DAC 2025

## PRELIMINARY ROUND



**Data Analysis  
Competition 2025**

**Team Name : Heisenberg**

**Himpunan Mahasiswa Statistika ITS (HIMASTA-ITS)**

Gedung H Lantai III, Jl. Arief Rahman Hakim  
Kampus ITS Sukolilo Surabaya

Email: pekanrayastatistika2025@gmail.com

Contact Person:

Hita (+62 857-1207-7572)

Ali (+62 812-3590-7687)

## CHAPTER I: Introduction

In recent years, the tech world has seen a surge in the adoption of large language-based artificial intelligence models, often referred to as Large Language Models (LLMs), such as GPT, Claude, Gemini, and other similar models. This development has been driven by theoretical advances in attention-based architectures, increased computing capacity, and massive scaling of training data, enabling these models to generate coherent text and perform a variety of language tasks at previously unimaginable levels of performance. It is this phenomenon of scaling and cross-task generalization that is driving the rapid integration of LLMs into commercial products and public services [1], [2], [3], [6].

As LLMs become more prevalent in everyday applications, from chatbots to writing assistance features, user interaction with these systems has increased. One of the most visible forms of interaction is user reviews on app distribution platforms and social media. These reviews appear unsolicited and often reflect users' subjective experiences with specific versions of products, new features, or service updates. Because they are direct and contextual, user reviews are an attractive source of data for developers who want to evaluate user responses in the real world [5].

User reviews can be used for text analysis, including sentiment analysis and opinion mining, in an effort to structure free-form and unstructured feedback into actionable insights. By examining patterns in reviews, organizations can identify features that are liked or disliked, recurring user experience patterns, and shifts in public perception of specific products or updates. This analysis has the potential to accelerate the product improvement cycle by providing empirical signals from the active user base [1], [2].

The use of reviews for text analysis faces a set of complex problems. Reviews are often unrepresentative reviewers tend to occupy the extremes of experience (either highly satisfied or highly dissatisfied) so aggregate summaries can be misleading. Moreover, review data are vulnerable to manipulation (e.g., fake reviews) and are shaped by platform moderation policies, which reduce the reliability of the signal if not audited carefully [5], [6]. From a linguistic perspective, reviews frequently exhibit informal writing styles, code-switching, and non-verbal digital elements such as emoji that carry distinct affective meaning; neglecting these elements risks discarding important dimensions of the underlying sentiment [9]. In addition, imbalanced sentiment distributions and fairness concerns create a need for evaluation criteria that go beyond aggregate accuracy to measure sensitivity to minority classes and under-represented user groups [10], [7].

To address these challenges, literature and best practices recommend multi-principled and governance-oriented approaches. First, treat data collection and preprocessing as critical stages this includes explicit handling of non-verbal elements (e.g., representing emoji), checking sample representativeness, and detecting anomalies or evidence of manipulation. Second, combine large-scale

quantitative analysis with qualitative verification so that signals produced at scale can be contextually validated before informing product decisions. Third, implement transparency and audit protocols (for example, dataset documentation and process audits) to ensure accountability and to minimize the risk of misuse of review data. Fourth, design evaluation metrics and intervention policies that incorporate cross-group fairness to avoid reinforcing unequal user experiences in product recommendations and engineering responses [8], [7].

In short, although the emergence of LLMs opens substantial opportunities to enhance language-based products, leveraging user reviews as a source of feedback requires a layered, careful analytical strategy, preprocessing techniques attuned to the data's characteristics, validation and audit procedures to preserve signal reliability, and governance principles that ensure insights are used responsibly and equitably.

## CHAPTER II: Theoretical Framework

### A. Sentiment Analysis

Sentiment analysis, also known as opinion mining, is a branch of Natural Language Processing (NLP) that focuses on identifying, extracting, and classifying opinions or emotions in text. The goal is to determine the polarity of a statement as positive, negative, or neutral in order to understand the author's attitude toward a particular topic [11], [12].

### B. TF-IDF

Term Frequency–Inverse Document Frequency (TF-IDF) is a statistical method used to evaluate the importance of a word in a document relative to the entire collection of documents (corpus). This concept combines the frequency of a word's appearance in a particular document (term frequency) with a measure of how rarely that word appears in all documents (inverse document frequency). A high TF-IDF value indicates that the word is frequently used in a particular document but rarely appears in other documents, making it a good marker for distinguishing that document [13], [14].

$$TF(t,d) = f(t,d) / \sum f(t',d) \quad (2.1)$$

$$IDF(t,D) = \log (N / (1 + |\{d \in D : t \in d\}|)) \quad (2.2)$$

$$TF-IDF(t,d,D) = TF(t,d) \times IDF(t,D) \quad (2.3)$$

dengan:

- $f(t,d)$  = number of occurrences of term  $t$  in document  $d$
- $\sum f(t',d)$  = total number of all terms in document  $d$

- $N$  = total number of documents in corpus  $D$
- $|\{d \in D : t \in d\}|$  = number of documents containing the term  $t$

TF measures the importance of a word in a single document, while IDF reduces the weight of words that appear too commonly in many documents. Thus, TF-IDF combines the two to assess the significance of words in a more balanced manner.

### C. Logistic regression

Logistic regression is a supervised machine learning classification algorithm used to predict the probability of a categorical target variable. Although simple, logistic regression is very efficient and effective, especially for binary and multi-class classification problems. Therefore, logistic regression is often used as a strong baseline model in text classification tasks [15].

$$P(y=1|x) = \sigma(z) = 1/(1+e^{(-z)}) \quad (2.4)$$

dengan:

- $z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$
- $\sigma(z)$  = sigmoid function
- $\beta$  = regression coefficient studied from
- $x$  = fitur input

### D. Model Transformer (XLM-RoBERTa)

Transformer is a deep learning architecture introduced by Vaswani et al. (2017) with a self-attention mechanism that allows the model to understand the context of words in a sentence more deeply [16]. RoBERTa (Robustly Optimized BERT Pretraining Approach) is an extension of BERT that has been optimized with larger data and better training strategies, resulting in higher performance on various NLP tasks [17]. In a multilingual context, XLM-RoBERTa (XLM-R) was developed to handle text from more than 100 languages, making it effective for multilingual text classification and sentiment analysis on mixed-language data.

### E. Weighted Loss

Weighted Loss is a technique used to address class imbalance in data. In situations where the amount of data in one class is much greater than in other classes, the model tends to be biased towards the majority class. Weighted Loss gives a higher weight to the minority class, so that classification errors in that class are given greater weight in the learning process [18]. The general formula for weighted cross-entropy loss can be written as follows:

$$L = - \sum w_i [y_i \log(\hat{y}_i)] \quad (2.5)$$



## F. Macro F1-Score

F1-Score is the harmonic mean of Precision and Recall, which is used to measure the performance of a classification model by balancing both metrics [19]. Macro F1-Score calculates F1 for each class independently, then takes the average, so that all classes have the same weight regardless of the data distribution. The F1-Score formula is:

$$\text{Macro F1} = (1 / N) \sum F1_i \quad (2.6)$$

Where N is the number of classes and  $F1_i$  is the F1 score for class i. This metric is very useful on imbalanced datasets because it is not affected by class size.

## CHAPTER III: Analytical Steps

### A. Data Collection and Preparation

The primary data source in this study is public user reviews for six leading Large Language Model (LLM) applications. The data was collected from various separate CSV files, with a total of 130,885 raw reviews. Each data row contains several attributes, including Comment (review text), At (review time), AppVersion (application version), Sentiment (class label: 0 for Negative, 1 for Neutral/Mixed, 2 for Positive), and source (name of the application where the review originated). After all the data was combined into a single DataFrame using the pandas library, essential data preparation steps were performed. First, data rows with empty Comment columns were deleted, as they did not contain any textual information that could be analyzed. Second, empty values in the AppVersion column were filled with the categorical value “unknown.” This decision was made to preserve valuable reviews so they could still be used in model training, without having to sacrifice data just because version information was unavailable.

### B. Pra-pemrosesan Data

Raw review text is highly heterogeneous and contains a lot of noise. To transform this data into a clean format that can be processed by the model, a series of pre-processing functions are applied sequentially to each review.

#### 1. Character Repetition Normalization

Characters that are repeated three or more times (for example, ‘a’ in ‘awesome’) are standardized to only two times (‘awesom<sup>2</sup>e’). This is done using the regular expression  $(.)\{2,\}$  to reduce insignificant word variations while still retaining a slight emphasis signal from the user. For example, “loveee it” will change to “lovee it”.

#### 2. Emoji Translation

Setiap karakter emoji diubah menjadi representasi teks deskriptif standarnya (CLDR *short name*) menggunakan pustaka emoji. Langkah ini krusial untuk



Data Analysis  
Competition 2025

# DATA ANALYSIS COMPETITION

INSTITUT TEKNOLOGI SEPULUH NOPEMBER



mengekstrak makna sentimen dari emoji, yang merupakan sinyal penting dalam ulasan informal. Contohnya "Sangat membantu 👍" akan diubah menjadi "Sangat membantu :thumbs\_up:".

### 3. Case Folding

All text is converted to lowercase to ensure consistency and prevent the model from treating the same word as different entities (e.g., "Good" and "good").

### 4. Cleansing

Characters other than letters (a-z), numbers (0-9), spaces, and symbols relevant to emoji conversion (👍 and :) are removed. This is to eliminate punctuation and other symbols that may interfere with the modeling process.

In this study, general steps such as Stopword Removal and Stemming were deliberately omitted. This decision was based on the fact that modern Transformer models such as XLM-RoBERTa are capable of capturing the contextual nuances of entire sentences. Removing stopwords (e.g., "not") or changing words to their base form risks eliminating crucial information that the model needs to understand meaning accurately.

## C. Experimental Design and Data Distribution

To objectively evaluate the model's performance, the cleaned dataset was divided into 80% training data and 20% test/validation data using the `train_test_split` function from `scikit-learn`. This split uses the `stratify` parameter on the `Sentiment` column. This ensures that the class distribution (the proportion of Positive, Negative, and Neutral reviews) remains the same in both data sets, thereby preventing bias in the evaluation and improving the reliability of the results. The use of `random_state=42` ensures that this data split can be replicated consistently.

## D. Modeling and Classification

Two fundamentally different modeling approaches were implemented and trained on the same training data for performance comparison.

### 1. Baseline Model: TF-IDF and Logistic Regression

The baseline approach in this study represents a classic machine learning method. The cleaned review text is first converted into a numerical representation using `TfidfVectorizer`. This vectorization process is configured to capture single words and two-word phrases (n-gram 1,2) by limiting the feature dimension to the top 20,000 terms. The resulting TF-IDF vectors then become the input for the Logistic Regression classification model. To directly address the class imbalance issue at this stage, the `class_weight='balanced'` parameter is enabled, which automatically adjusts the class weights inversely proportional to their frequency.

### 2. Advanced Model: Fine-Tuning XLM-RoBERTa

The second approach utilizes the power of advanced deep learning architecture through the XLM-RoBERTa model. Clean review text first goes through a tokenization process using the appropriate `AutoTokenizer`, where each review is converted into a sequence of tokens with a uniform length of 128 through

a padding and truncation process. The pre-trained XLM-RoBERTa model is then fine-tuned on the tokenized data. To specifically address the class imbalance issue, a special CustomTrainer class was created to implement a weighted loss function strategy. This mechanism works by applying CrossEntropyLoss that has been modified with class weights, so that the model is given a higher penalty when it makes classification errors on minority sentiment classes during the training process.

### E. Model Performance Evaluation

The final evaluation of both models was performed on 20% of the test data that was never used during training. The main metric used for comparison was the Macro F1-Score. The F1-Score is calculated as the harmonic mean of Precision and Recall, providing a balanced measure of the model's ability to avoid mislabeling (Precision) and its ability to find all relevant samples (Recall). The “Macro” approach calculates the F1-Score for each class separately and then takes the arithmetic mean. This gives equal weight to each class, making it a very reliable and fair metric for evaluating performance on datasets with highly imbalanced class distributions, such as in this study.

## CHAPTER IV: Analysis of Results

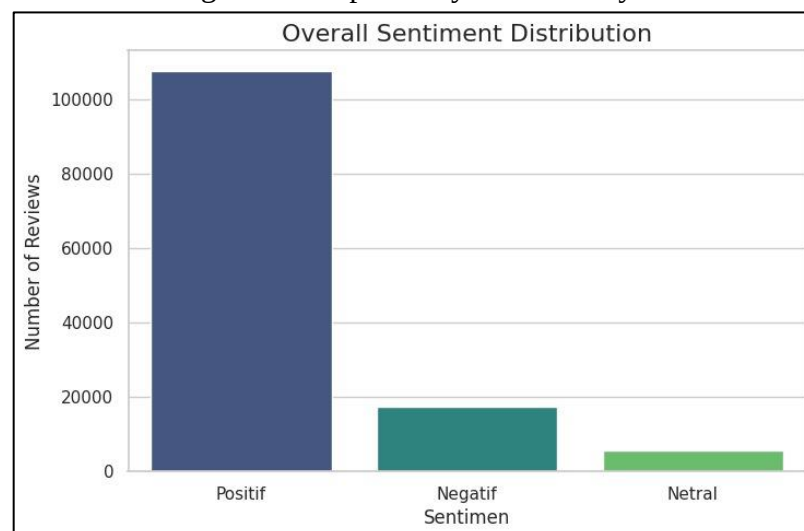
### A. Section 1: Sentiment Classification Modeling

This section focuses on the process of building, evaluating, and interpreting sentiment classification models.

#### 1. Exploratory Data Analysis (EDA)

Prior to modeling, exploratory data analysis was performed on the cleaned dataset to understand its main characteristics. The results of the analysis showed extreme class imbalance, where the Positive sentiment class (2) comprised approximately 82.3% of the total data, followed by the Negative class (0) at 13.4%, and the Neutral/Mixed class (1) at only 4.3%.

Figure 4.1 Exploratory Data Analysis



As illustrated in Figure 4.1, the research dataset shows a significant class imbalance, where reviews with positive sentiment (82.3%) greatly dominate the class distribution.

## 2. Model Development and Evaluation

Two main models were developed and evaluated on the validation dataset (20% of the total data) using Macro F1-Score as the main metric. The first approach, which served as the baseline model, used a combination of TF-IDF features (with n-grams of 1 and 2) and Logistic Regression classification, equipped with the parameter `class_weight='balanced'` to handle class imbalance. In comparison, the second, more advanced approach is the Transformer XLM-RoBERTa model. This model was fine-tuned for 3 epochs and implemented a weighted loss strategy through a CustomTrainer to specifically address the issue of data imbalance more effectively. The comparative performance results of the two models developed are then presented in detail in Table 2.1.

Table 2.1 Performance Results of the Comparative Model on the Validation Set

| Model       | Macro F1-Score | Accuracy |
|-------------|----------------|----------|
| Baseline    | 0.62           | 0.89     |
| Transformer | 0.65           | 0.91     |

## 3. Interpretation of Modeling Results

From the evaluation results, it is clear that the Transformer XLM-RoBERTa model with the weighted loss strategy shows far superior performance. This model achieved a Macro F1-Score of 0.6550 on the local validation set, significantly surpassing the baseline model (0.62). This improvement in performance is driven by the effectiveness of the weighted loss strategy in forcing the model to better recognize minority classes (Negative and Neutral), which was previously a major weakness. This proves that the Transformer model's understanding of context, combined with weight penalties in the loss function, is very effective in recognizing ambiguous and often overlooked reviews.

Furthermore, the consistency between the local validation score (0.6550) and the score on the Kaggle public leaderboard (0.64802) shows that the model has good generalization capabilities and does not overfit on the training data. This provides a high degree of confidence that the model will perform well on completely new data.

## CHAPTER V: Conclusion and Recommendation

### A. Conclusion

This study has successfully developed and evaluated a pipeline for sentiment analysis on multilingual and unbalanced LLM application user reviews. Based on a systematic comparison of models, several key conclusions can be drawn: feature





Data Analysis  
Competition 2025

# DATA ANALYSIS COMPETITION

INSTITUT TEKNOLOGI SEPULUH NOPEMBER



importance, and discuss how well the model generalizes across multiple LLM platforms.

1. The Transformer model (XLM-RoBERTa) optimized with a weighted loss strategy proved to be the most superior approach. This model achieved a Macro F1-Score of 0.6550 on the local validation set, significantly surpassing the performance of the baseline model (TF-IDF and Logistic Regression) which scored 0.62.
2. Class imbalance is the most critical challenge in this dataset. The standard Transformer model, despite its high accuracy, shows a lower F1-Score performance than the baseline due to its failure to recognize minority classes. This proves that the application of explicit imbalance handling strategies, such as weighted loss, is an absolutely necessary step to obtain reliable performance evaluations and functional models.
3. Analysis of negative reviews classified by the best model successfully identified recurring user complaint themes, such as performance issues (speed), answer accuracy (hallucinations), and feature limitations. In addition, sentiment analysis across application versions shows a strong correlation between new version releases and fluctuations in user sentiment, which can be used to track the impact of bugs or feature launches.

## B. Recommendation

Based on the above conclusions, several practical recommendations can be formulated for LLM application developers:

1. For Product and Development Teams: It is recommended to adopt an automated sentiment monitoring system that uses Macro F1-Score as the primary metric, rather than just accuracy. Identified complaint themes (performance, accuracy, etc.) should be used as direct input to prioritize bug fixes and feature development in the next product roadmap.
2. For User Experience (UX) and Marketing Teams: Sentiment analysis per application version should be integrated as part of the release cycle. By tracking sentiment after each update, teams can quickly gauge market acceptance of changes (e.g., new interface design) and proactively identify regressions or declines in user satisfaction.

### Himpunan Mahasiswa Statistika ITS (HIMASTA-ITS)

Gedung H Lantai III, Jl. Arief Rahman Hakim

Kampus ITS Sukolilo Surabaya

Email: pekanrayastatistika2025@gmail.com

Contact Person:

Hita (+62 857-1207-7572)

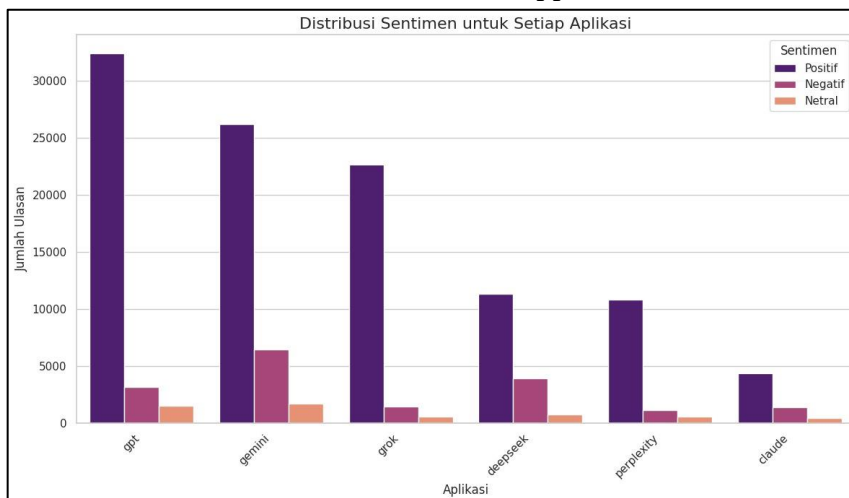
Ali (+62 812-3590-7687)

## REFERENCES

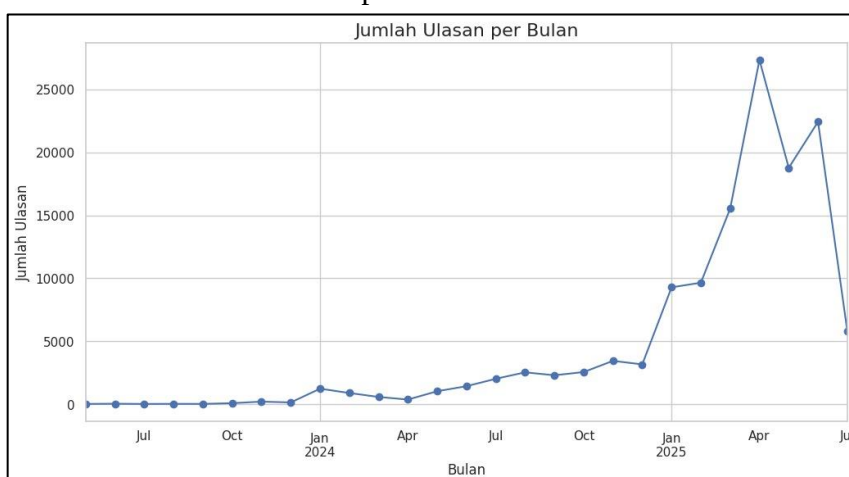
- [1] A. Vaswani et al., "Attention Is All You Need," in Advances in Neural Information Processing Systems (NeurIPS), 2017.
- [2] T. B. Brown et al., "Language Models are Few-Shot Learners," in Proc. NeurIPS, 2020.
- [3] J. Kaplan et al., "Scaling Laws for Neural Language Models," arXiv:2001.08361, 2020.
- [5] M. A. Lazer et al., "The science of fake news," Science, vol. 359, no. 6380, pp. 1094–1096, 2018.
- [6] E. M. Bender, T. Gebru, A. McMillan-Major, and S. Shmitchell, "On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?," in Proc. ACM FAccT, 2021.
- [7] J. Buolamwini and T. Gebru, "Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification," in Proc. FAT '18, 2018.
- [8] T. Gebru et al., "Datasheets for Datasets," arXiv:1803.09010, 2018.
- [9] M. Novak, J. Smailović, B. Sluban, and I. Mozetič, "Sentiment of Emojis," PLOS ONE, vol. 10, no. 12, e0144296, 2015.
- [10] X. He and Y. Garcia, "Learning from Imbalanced Data," IEEE Trans. Knowl. Data Eng., vol. 21, no. 9, pp. 1263–1284, 2009.
- [11] B. Pang and L. Lee, "Opinion mining and sentiment analysis," Foundations and Trends in Information Retrieval, vol. 2, no. 1–2, pp. 1–135, 2008.
- [12] B. Liu, Sentiment Analysis and Opinion Mining. San Rafael, CA: Morgan & Claypool Publishers, 2012.
- [13] G. Salton and C. Buckley, "Term-weighting approaches in automatic text retrieval," Information Processing & Management, vol. 24, no. 5, pp. 513–523, 1988.
- [14] C. D. Manning, P. Raghavan, and H. Schütze, Introduction to Information Retrieval. Cambridge: Cambridge University Press, 2008.
- [15] D. W. Hosmer, S. Lemeshow, and R. X. Sturdivant, "Applied Logistic Regression," 3rd ed., Wiley, 201
- [1] A. Vaswani et al., "Attention Is All You Need," Advances in Neural Information Processing Systems, 2017.
- [2] Y. Liu et al., "RoBERTa: A Robustly Optimized BERT Pretraining Approach," arXiv preprint arXiv:1907.11692, 2019.
- [3] H. He and E. A. Garcia, "Learning from Imbalanced Data," IEEE Transactions on Knowledge and Data Engineering, 2009.
- [4] C. J. van Rijsbergen, "Information Retrieval," Butterworths, 1979.

## ATTACHMENT

### Attachment 1: Sentiment Distribution for Each Application



### Attachment 2: Number of Reviews per Month





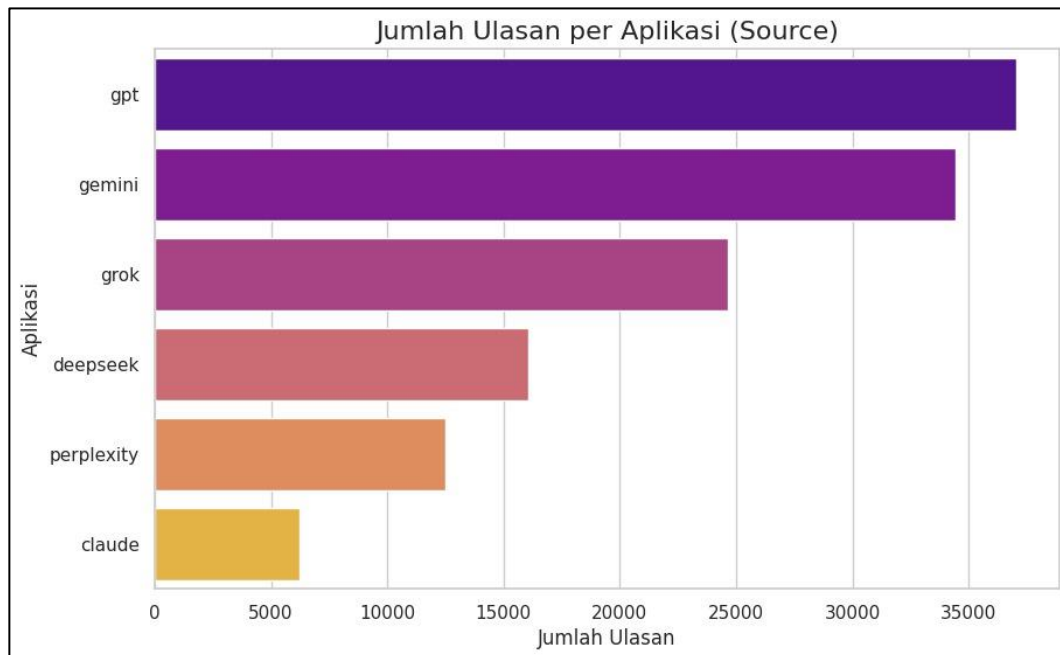
Data Analysis  
Competition 2025

# DATA ANALYSIS COMPETITION

INSTITUT TEKNOLOGI SEPULUH NOPEMBER



## Attachment 3: Number of Reviews per Application (Source)



### Himpunan Mahasiswa Statistika ITS (HIMASTA-ITS)

Gedung H Lantai III, Jl. Arief Rahman Hakim  
Kampus ITS Sukolilo Surabaya

Email: pekanrayastatistika2025@gmail.com

Contact Person:

Hita (+62 857-1207-7572)

Ali (+62 812-3590-7687)